Code No: 121AB **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year Examinations, March/April - 2023
MATHEMATICS - I
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, AME, MIE, PTM)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a) Define Hermitian, Skew-Hermitian and Unitary Matrices. [2]
b) Define the rank and nature of a quadratic form. [3]
c) Write the geometrical interpretation of Lagrange's mean value theorem. [2]
d) Explain in brief the method of Lagrange multipliers. [3]
e) Show that $\beta(m,n)=\beta(n,m)$. [2]
f) Evaluate $\int_0^1\int_1^2\int_2^3 xy^2z^3\,dxdydz$. [3]
g) Write the differential equation of the family of orthogonal trajectories of the curves $xy=c^2$. [2]
h) Find the general solution of $y''+2y'+y=0$. [3]
i) Write the linearity property in Laplace transforms. [2]
j) Define a periodic function. What is the Laplace transform of a periodic function? [3]

PART - B

(50 Marks)

2.a) Find the rank of the matrix $A=\begin{pmatrix}1&2&2&4\\2&3&4&6\\3&5&6&10\\-1&1&-2&-2\end{pmatrix}$.
b) Solve completely, the system of equations $x+y-2z+3w=0;\ x-2y+z-w=0;$ $4x+y-5z+8w=0;\ 5x-7y+2z-w=0.$ [5+5]

OR

3. Verify Cayley-Hamilton theorem for the matrix $A=\begin{pmatrix}1&2&-1\\2&1&-2\\2&-2&1\end{pmatrix}$ and hence find its inverse. [10]

4. State Lagrangers mean value theorem, and using it prove that when $0 < a < b < 1$, $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$. Hence show that $\frac{\pi}{4} + \frac{3}{25} < \tan^{-1}\frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$. [10]

OR

5. Find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. [10]

6.a) Express the following integrals in terms of Gamma function:

i) $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$ ii) $\int_0^{\infty} a^{-bx^2} dx$

b) Find $\Gamma\left(-\frac{11}{2}\right)$. [5+5]

OR

7.a) By changing into polar coordinates, evaluate $\int_0^{\infty}\int_0^{\infty} e^{-(x^2+y^2)} dxdy$.

b) Using triple integral, find the volume of a sphere of radius 'r'. [5+5]

8.a) Solve $xy(1+xy^2)\frac{dy}{dx} = 1$.

b) If the temperature of the air is 30^0C, and the substance cools from 100^0C to 80^0C in 10 minutes, find the temperature of the substance after 20 minutes. [5+5]

OR

9. Solve $(D-2)^2 y = 8(e^{2x} + \sin 2x + x^2)$. [10]

10. Find the Laplace transform of the function:

a) $f(t) = 2e^{-3t}\sin t + 3e^t \sin 4t + 7\sqrt{t}$

b) $\int_0^t \frac{\sin t}{t} dt$ [5+5]

OR

11. Solve by Laplace transform technique $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = \sin t, y = \frac{dy}{dt} = 0$ *when* $t = 0$. [10]

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